INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

DOUGH MIXING MACHINE (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers a two-speed, horizontal-type machine for mixing bread dough for Naval shipboard use.

2. CLASSIFICATION

Size 1: 150 pounds finished dough capacity
Size 2: 300 pounds finished dough capacity
Size 3: 600 pounds finished dough capacity

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The machine shall consist of a frame supporting a mixing bowl, in which a rotating agitator is driven from a motor by means of V-belts, silent or roller chain, gearing, or a combination of such driving means. The machine shall be of rigid construction and shall be capable of thoroughly developing bread dough from the ingredients listed in Armed Forces Recipe Service (AFRS) D-G-2, without splashover of ingredients or overheating of motors and bearings. The machine shall thoroughly mix dough without slicing or other outside assistance, and shall leave no dry flour in the tank.

3.2 Height. Size 1 and size 2 machines shall have a height not greater than 65 inches. Size 3 machines shall have a height not greater than 70 inches.

3.3 Stainless steel materials. Stainless steel shall be in accordance with ASTM A167, type 302 or 304.

3.4 Frame. The frame shall be of steel suitable for supporting the mixing bowl when operated at full load. The base of the frame shall be a sanitary wash-under type.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 7320

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3.5 Mixing bowl. The mixing bowl (bowl sheet, bowl ends, and bowl cover) shall be constructed of stainless steel sheets or plates. The thickness of the bowl sheet and cover shall be not less than 25-gauge. The thickness of the bowl ends shall be not less than 16-gauge. A small fillet of stainless steel shall be continuously welded, ground smooth, and coved to a radius of not less than 0.125 inch around the entire juncture of bowl sheet and bowl ends. Trunnions shall be made of steel or ductile iron and shall be welded, bolted, or riveted to both bowl ends. When tilted to the maximum position, the bowl shall permit emptying the dough into a trough 26 inches high. The mixing bowl cover shall be provided with a flour inlet gate not less than 12 inches in diameter, and with an adequate water inlet. The bowl shall be provided with an indicating thermometer, adequately protected, at a convenient height on the front of the bowl for observation. The thermometer shall indicate the temperature of the dough within the bowl.

3.6 Agitating device. The agitating device shall be of the single arm nonreversing type with shafts extending through the ends of the mixing bowl. The agitating device and shafts shall be made of stainless steel. Either one or both shafts shall be provided on the outer end with a driving means (such as a gear or sprocket) for connection to motive power. The agitator shall have two speeds, operating in the range of 20 to 22 revolutions per minute for low speed, and in the range of 40 to 44 revolutions per minute for high speed, obtained by means of a two-speed electric motor.

3.7 Time clocks. Two time clocks with splashproof faces shall be mounted on the front of each machine. The clocks shall automatically switch the mixer from low speed to high speed when the predetermined low speed time has elapsed, and automatically shut the mixer off when the full predetermined mixing time (low speed time plus high speed time) has elapsed.

3.8 Safety switch. The machine shall be provided with a safety switch controlled by a JOG button so that the operator, using both hands, must simultaneously press the JOG button and the START button to rotate the agitator when the bowl is tilted 6 inches or more from the vertical or operating position. The safety JOG button and the START button shall be located not less than 10 inches apart.

3.8.1 Emergency stop switch. The machine shall be provided with a safety feature that automatically shuts down the machine when the hood is lifted.

3.9 Dumping device. Devices for dumping or emptying the bowl shall be provided as follows.

3.9.1 Sizes 1 and 2 machines. A manually operated device shall be provided, incorporating a worm gear set, arranged so that the bowl is at all times automatically locked in any given position.

3.9.2 Size 3 machines. A power dump shall be provided, complete with an adequately powered reversible brake type motor; controls and push button stations labeled UP, DOWN and STOP; and a safety switch, including a safety JOG button, to prevent the bowl from being closed beyond four-fifths of the total opening (unless the operator maintains a two-hand contact to cause the tilting motor to complete the bowl closure).

3.10 Hood. A stainless steel hood shall cover the top of the machine, and shall be supported by the frame of the machine. The front portion of the hood shall be hinged.

3.11 Housing. The machine shall be completely enclosed and provided with safety guards so that moving parts (such as gears and chains) do not present a

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safety hazard. Safety guards shall be readily removable for cleaning and maintenance.

3.12 Mounting. The unit shall be mounted on four 6-inch high stainless steel legs suitable for secure attachment to the deck.

3.13 Shafts. Shafts, except agitating device shafts, shall be high grade, cold-rolled or cold-drawn machinery steel, and shall run in ball bearings or bearings having bronze or other bushings appropriate for their function.

3.14 Lubrication. Pressure-gun lubrication fittings shall be provided at readily accessible points for all bearings requiring grease lubrication. Gear boxes, where used, shall be provided with accessible screw plugs for filling and draining. Lubricants of all types shall be prevented from entering the mixing bowl.

3.15 Agitator shaft seals. Shaft seals shall be provided at each end of the bowl and shall prevent any leakage of the dough or ingredients at these points. Shaft seals shall be independent of the dough or ingredients at these points. Shaft seals shall be independent of the agitator shaft bearings and shall not carry any part of the agitator load. Shaft seals shall be accessible and removable for cleaning and renewal of the seal rings.

3.16 Gear or chain drive. All pinions and gears shall be steel and shall have cut teeth. The cutting of the gears, alignment of the shafts and gears, and deflection of these shafts shall be such that the error in action shall be not greater than 0.002 inch. Spiral bevel gears shall be mounted to reduce the deflection and endwise displacement of the gears under load. Spiral bevel gears shall be mounted on either ball bearings or taper roller bearings to fully provide for both thrust and radial loads. Means shall be provided for accurately positioning the spiral bevel gears on their shafts. The gears shall be keyed to the shaft and means shall be provided to prevent linear movements along the shaft. Nuts which tighten whenever the machine starts shall be used for holding gears on shafts, otherwise lockwashers and lock nuts shall be used. The fit of the gears on their shafts or bearings shall be in accordance with AGMA 2000, Class 3 fit or better. All gears, sprockets, and chains shall be lubricated and enclosed in chambers sealed against loss of lubricant. Gears and chains shall be readily accessible for inspection. Studs and shafts for mounting bearings are not acceptable unless the elongation in a 2-inch sample of the metal is not less than 10 percent. Means shall be provided for adjustment of chain tension.

3.17 V-belt. Belts, when used, shall be V-type, provided in accordance with the belt manufacturer's recommendations based on load ratings, with adequate safety factors, greater than the maximum belt loads resulting under extreme operating conditions. The V-belts shall have a service life of not less than 5000 hours of operation.

3.18 Electrical requirements. The unit shall operate on 440 VAC, 60 Hz, three-phase power, as defined in DOD-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (3 conductors) and for equipment grounding (1 conductor). All outermost surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be no greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 6.2)

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3.19 Electric motors. The motors shall be in accordance with NEMA MG1. The motors shall function under all operating conditions without exceeding permissible temperature rise. In addition, main drive motors shall have the following characteristics:

Service	C
Bearings	Ball or sleeve
Ambient temperature	50 °C
Enclosure	Open and dripproof
Duty	Continuous
Cooling	Natural or fan
Horsepower	To suit load
Speed	Two-speed, constant for both windings
Insulation	Class B, or better

3.20 Shock hazard labels. A label reading "Danger - Shock Hazard" shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger - Shock Hazard.
- Power supply must be disconnected before servicing.
- Access covers must be in place before use.
- Service should be performed by authorized personnel only.

3.21 Controllers. Contactors shall be used for all power circuits and shall switch all three lines of three phase circuits and both sides of single phase circuits. Control circuits shall operate at not greater than 115 VAC. Solenoid valves shall have epoxy encapsulated coils.

3.22 Inclined operation. The unit shall operate satisfactorily in accordance with the requirements of this CID (such as hood remaining closed to prevent loss of contents) with no spillage of liquid or product when test-operated for 30 seconds inclined at an angle of 15° each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

3.23 Physical size limitations. The machine shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8-inch radius corners. (Shipboard requirement, see 7.2)

3.24 Finish. The unit shall have the manufacturer's standard commercial exterior finish in accordance with NSF standards.

3.25 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 8 and UL 969.

3.26 Label plates. The unit shall be provided with a data nameplate and an instruction plate. They shall be attached to the unit, readily visible during normal operating use, and shall not adversely affect the life and utility of the unit.

3.26.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

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3.26.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Size
- When required, manuals shall be in accordance with ASTM F760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Sources of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

DOD-STD-1399	-	Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment

7.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A167	-	Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM D3951	-	Commercial Packaging
ASTM F760	-	Food Service Equipment Manuals

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7.3.3 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 8 - Commercial Powered Food Preparation Equipment

7.3.4 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 969 - Marking and Labeling Systems

7.3.5 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.3.6 The National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from The National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA MG1 - Motors and Generators

7.3.7 The Association of Gear Manufacturers (AGMA) Standards. AGMA Standards are available from the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, VA 22314-2730.

AGMA 2000 - Gear Classification and Inspection Handbook Tolerances and Measuring Methods for Unassembled Spur and Helical Gears

7.4 Suggested sources of supply. One manufacturer known to meet the requirements of this CID is listed below. However, source of supply is not limited to this company, if others are found to meet the requirements of this CID.

Triumph Manufacturing Co.
11180 Southland Road
P.O. Box 40810
Cincinnati, OH 45240

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

Preparing activity:
Navy - SH
(Project 7320-0965)

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